WORKED SOLUTION

STEP I 2010, Question 6

Pure • Reduction of Order

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s1-q6

The key idea

Once we know one solution of a linear second-order equation, writing the general solution as that known solution times an unknown function u makes every term in u itself (with no derivative) cancel. What is left involves only u' and u'', so it is really a *first*-order equation for $v = u'$, and this one is separable. This technique is called *reduction of order*.

Solution

Checking that $y = \mathrm{e}^x$ works

If $y = \mathrm{e}^x$ then $y' = y'' = \mathrm{e}^x$, so

$$(x-1)\mathrm{e}^x - x\mathrm{e}^x + \mathrm{e}^x = (x - 1 - x + 1)\mathrm{e}^x = 0,$$

and $y = \mathrm{e}^x$ satisfies $(*)$.

The equation for u

Let $y = u\mathrm{e}^x$. By the product rule,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = (u' + u)\,\mathrm{e}^x,$$
$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = (u'' + u')\,\mathrm{e}^x + (u' + u)\,\mathrm{e}^x = (u'' + 2u' + u)\,\mathrm{e}^x.$$

Substituting into the left-hand side of $(*)$ and taking out the factor e^x:

$$\begin{aligned}&\mathrm{e}^x[(x-1)(u'' + 2u' + u) - x(u' + u) + u]\\ &\quad = \mathrm{e}^x[(x-1)u'' + (2x - 2 - x)u' + (x - 1 - x + 1)u]\\ &\quad = \mathrm{e}^x[(x-1)u'' + (x-2)u'].\end{aligned}$$

The terms in u cancel, exactly as they did when we checked $y = \mathrm{e}^x$ (that check is the case $u = 1$). Since e^x is never zero, $y = u\mathrm{e}^x$ satisfies $(*)$ if and only if

$$(x-1)\frac{\mathrm{d}^2u}{\mathrm{d}x^2} + (x-2)\frac{\mathrm{d}u}{\mathrm{d}x} = 0,$$

which is $(**)$, as required.

Solving for v and then u

With $v = \dfrac{\mathrm{d}u}{\mathrm{d}x}$, $(**)$ becomes the first-order equation

$$(x-1)\frac{\mathrm{d}v}{\mathrm{d}x} = -(x-2)\,v.$$

One solution is $v = 0$. Otherwise, working on an interval where $x \neq 1$ and $v \neq 0$, we separate the variables:

$$\int \frac{1}{v}\,\mathrm{d}v = -\int \frac{x-2}{x-1}\,\mathrm{d}x.$$

Ask yourself How do we integrate $\dfrac{x-2}{x-1}$?

Write the numerator in terms of the denominator: $x - 2 = (x-1) - 1$, so

$$\frac{x-2}{x-1} = 1 - \frac{1}{x-1}.$$

Hence

$$\ln|v| = -x + \ln|x-1| + c, \qquad \text{so} \qquad v = C(x-1)\mathrm{e}^{-x},$$

where $C = \pm\mathrm{e}^c$. Allowing $C = 0$ includes the solution $v = 0$ as well, so $v = C(x-1)\mathrm{e}^{-x}$ for an arbitrary constant C. (It is easy to check directly that this satisfies $(x-1)v' + (x-2)v = 0$ for all x, including $x = 1$.)

Now integrate to find u. Integrating by parts, with $(x-1)$ differentiated and e^{-x} integrated,

$$\begin{aligned}\int (x-1)\mathrm{e}^{-x}\,\mathrm{d}x &= -(x-1)\mathrm{e}^{-x} + \int \mathrm{e}^{-x}\,\mathrm{d}x \\ &= -(x-1)\mathrm{e}^{-x} - \mathrm{e}^{-x} = -x\mathrm{e}^{-x}.\end{aligned}$$

So $u = -Cx\mathrm{e}^{-x} + B$. Renaming $-C$ as A,

$$u = Ax\mathrm{e}^{-x} + B,$$

where A and B are arbitrary constants.

The final part

We showed above that $y = ue^x$ satisfies $(*)$ whenever u satisfies $(**)$. Our $u = Axe^{-x} + B$ satisfies $(**)$, because its derivative is $v = -A(x-1)e^{-x}$, which solves the equation for v. Therefore

$$y = ue^x = (Axe^{-x} + B)e^x = Ax + Be^x$$

satisfies $(*)$ for any constants A and B, as required.

Checks

Check $y = x$ directly: $y' = 1$, $y'' = 0$, so the left-hand side of $(*)$ is $0 - x + x = 0$. Since $(*)$ is linear, any combination $Ax + Be^x$ of the two solutions x and e^x also works. As a numerical check, with $A = 2$, $B = -1$ and $x = 0.5$ we have $y = 1 - e^{0.5} \approx -0.64872$, $y' = 2 - e^{0.5} \approx 0.35128$ and $y'' = -e^{0.5} \approx -1.64872$, giving $(-0.5)(-1.64872) - 0.5(0.35128) - 0.64872 = 0.82436 - 0.17564 - 0.64872 = 0$.

Common slips

- Forgetting that the product rule gives $2u'$ (not u') in y''.
- Integrating $\frac{x-2}{x-1}$ as a logarithm straight away, without first splitting off the 1.
- Dropping the constant when integrating v to get u. That constant B is what brings back the original solution e^x.

Beyond STEP

Reduction of order works for any linear equation $y'' + p(x)y' + q(x)y = 0$: if y_1 is one solution, putting $y = uy_1$ always gives a first-order equation for u'. Together with Abel's formula for the Wronskian, it shows that the solutions form a two-dimensional vector space, so $Ax + Be^x$ is in fact the general solution here, on each interval not containing the singular point $x = 1$. Boyce and DiPrima, *Elementary Differential Equations*, covers both ideas.

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